PINCH MOVES AND THE NONORIENTABLE FOUR-GENUS OF KNOTS WITH AT MOST THIRTEEN CROSSINGS

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ABSTRACT. The smooth nonorientable four-genus $\gamma_4(K)$ of a knot is the least first Betti number of a smooth nonorientable surface in B^4 bounded by K. KnotInfo leaves it undetermined for 3525 prime knots with eleven to thirteen crossings. We determine it for 1954 of them and lower the known upper bound for 578 more. Each result comes from one pinch move, the nonorientable band move at a crossing of a diagram, that turns the knot into a slice knot or into a knot whose nonorientable four-genus is known. Every certificate is a diagram, a crossing and a target knot. A program independent of the search checks each one by Reidemeister moves or by an isomorphism of triangulated knot complements.

1. INTRODUCTION

Every knot $K \subset S^3 = \partial B^4$ bounds a smooth surface in B^4; if K is not slice, it bounds nonorientable ones as well. The *nonorientable four-genus* $\gamma_4(K)$ is the least first Betti number of a smooth compact nonorientable surface $F \subset B^4$ with $\partial F = K$; for slice knots one sets $\gamma_4(K) = 1$. It was studied by Murakami and Yasuhara [9] and by Gilmer and Livingston [4]. Its values on small knots were computed by Jabuka and Kelly for eight and nine crossings [6], by Ghanbarian for ten [3] and by Fairchild for nonalternating knots with eleven [2]. Lee and Sabloff [8] extended the computation to fourteen crossings with an algorithmic search for Möbius bands in the style of Dunfield and Gong [1], together with the Arf invariant obstruction of Yasuhara [12] and a linking form obstruction of Murakami and Yasuhara [9]. The KnotInfo database [7] incorporates these data. In its version 2026.10.5, 3525 knots with eleven to thirteen crossings have an interval $[a, b]$ with $a < b$ as the entry for γ_4.

Theorem 1. *For each of the 1954 knots listed in the data file* `certificates.json` *with the flag* `closed`*, $\gamma_4(K)$ equals the lower end of its KnotInfo interval: 389 knots have $\gamma_4(K) = 1$ and 1565 have $\gamma_4(K) = 2$. For the other 578 knots of the file the upper bound is lowered by at least one.*

Table 1 gives the counts by crossing number and by KnotInfo interval. The proof uses one elementary move.

Definition 2. Let D be a diagram of a knot K and c a crossing of D. Of the two smoothings of c, one respects the orientation of K and gives a two-component link; the other gives a knot K'. We say K' is obtained from K by a *pinch move* at c.

Lemma 3. *If K' is obtained from K by a pinch move, then $\gamma_4(K) \leq 1$ when K' is slice, and $\gamma_4(K) \leq \gamma_4(K') + 1$ in general.*

Proof. The pinch move is a band move, so there is a cobordism $\Sigma \subset S^3 \times [0, 1]$ from K to K' formed by $K \times [0, 1]$ and one band. Since K' is connected, the band does not respect the orientation of K, and Σ is a Möbius band with an open disk removed. If K' bounds a disk Δ in B^4, then $\Sigma \cup \Delta$ is a Möbius band bounded by K, so $\gamma_4(K) \leq 1$. If K' bounds a surface F with

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$b_1(F) = \gamma_4(K')$, then $\Sigma \cup F$ is a nonorientable surface bounded by K with $b_1 = b_1(F) + 1$. This is the bound used by Jabuka and Kelly [6]. □

The lower bounds are not ours. A knot that is not slice has $\gamma_4 \geq 1$. For the knots whose interval starts at 2, the lower bound is the obstruction to bounding a Möbius band recorded by KnotInfo from Lee and Sabloff [8]: either Yasuhara's condition $4\,\mathrm{Arf}(K) - \sigma(K) \in \{0, \pm 2\}$ (mod 8) fails [12], or the linking form obstruction of Murakami and Yasuhara [9] applies. Theorem 1 therefore rests on these published obstructions, on the slice status and four-genus values recorded in KnotInfo for the targets, and on the certificates below.

2. Certificates and their verification

A certificate for a knot K consists of a planar diagram code D, a crossing c of D, a choice of smoothing, and a target: either the unknot or a knot T of KnotInfo. It asserts that

(1) D is a diagram of K;
(2) the chosen smoothing of c gives a one-component diagram D';
(3) D' represents T or its mirror image;
(4) the bound $\beta = 1$ if T is slice or the unknot, and $\beta = \gamma_4(T) + 1$ otherwise, is smaller than the upper end of the KnotInfo interval of K.

Since γ_4 and sliceness are invariant under mirror image, (1) to (4) and Lemma 3 give $\gamma_4(K) \leq \beta$.

The certificates were found with SnapPy [11]: for every open knot we applied every pinch move to the KnotInfo diagram and to up to twenty-five diagrams obtained from it by random Reidemeister moves, and identified the resulting knots by hyperbolic structure. A second pass used the knots settled in the first pass as targets; the number of certificates with such a target is 3, and the verifier checks them after the certificates of their targets.

The verifier `verify.py` does not use SnapPy. It reads the KnotInfo database and uses Regina [10] for the topology. For (1) and (3) it proves that two diagrams represent the same knot up to mirror image in one of two ways.

- *Isotopy.* Both diagrams are simplified by Reidemeister moves to the same number of crossings, and the closures of the two minimal diagrams under Reidemeister moves that add at most one crossing share a diagram. Diagrams are compared by Regina's canonical signatures, taken up to reflection, rotation and reversal.
- *Complements.* The triangulated complements of the two diagrams have combinatorially isomorphic triangulations, possibly after Pachner moves that add at most one tetrahedron. Then the complements are homeomorphic, and by the theorem of Gordon and Luecke [5] the knots agree up to mirror image.

Both tests are exact: a positive answer is a proof. Regina's simplification is randomised, so the verifier makes up to four attempts, each from a fixed seed, and gives each certificate two minutes. For the unknot the verifier uses Regina's recognition of the solid torus. Item (2) is checked by following the strands of D', and (4) by reading KnotInfo.

The verifier accepts all 2532 certificates. It rejects certificates corrupted in four ways, on a sample of twenty: the other smoothing of the crossing, the next crossing, a wrong target, and a bound lowered by one (Table 2). The seven modified certificates with the next crossing that it accepts are valid certificates: that crossing gives the same target.

3. Results

The remaining 1571 knots, listed in `still_open.json`, are not settled by one pinch move found in this way. For a knot with interval $[1, 2]$ that is not settled, no pinch move we tried gives a slice knot; this says nothing about whether $\gamma_4 = 1$.

crossings	KnotInfo	knots	settled	lowered	open
11	[1, 2]	14	0	0	14
11	[1, 3]	9	0	9	0
11	[2, 3]	36	30	0	6
12	[1, 2]	87	25	0	62
12	[1, 3]	87	12	69	6
12	[2, 3]	278	237	0	41
12	[2, 4]	3	2	0	1
13	[1, 2]	890	225	0	665
13	[1, 3]	677	126	497	54
13	[1, 4]	3	1	2	0
13	[1, 5]	1	0	0	1
13	[2, 3]	1437	1294	0	143
13	[2, 4]	3	2	1	0
all		3525	1954	578	993

TABLE 1. Open entries of γ_4 in KnotInfo 2026.10.5 and what the certificates give. "Settled": the value is now known; "lowered": the upper bound is lower.

corruption	rejected
other smoothing of the crossing	20 of 20
next crossing of the diagram	13 of 20
wrong target ($12a_1$)	20 of 20
bound lowered by one	20 of 20

TABLE 2. Corrupted certificates rejected by the verifier.

Remark 4. The 389 Möbius bands found here are not in the data of Lee and Sabloff [8], whose search follows Dunfield and Gong [1] and runs over a finite family of bands chosen in advance. We have not determined which restriction of that family excludes them. The knots settled at $\gamma_4 = 2$ were outside their aim, which was Möbius bands; for them the target of the pinch move is a knot known to bound a Möbius band but not a disk.

APPENDIX A. THE SETTLED KNOTS

$\gamma_4 = 1$ **(389 knots).**

12a_133	12a_818	12n_147	13a_12	13a_343	13a_550
12a_200	12a_878	12n_382	13a_129	13a_353	13a_591
12a_212	12a_882	12n_390	13a_138	13a_365	13a_610
12a_218	12a_895	12n_444	13a_156	13a_369	13a_619
12a_313	12a_916	12n_495	13a_158	13a_375	13a_680
12a_390	12a_1069	12n_554	13a_166	13a_381	13a_712
12a_457	12a_1092	12n_555	13a_180	13a_392	13a_714
12a_474	12a_1094	12n_597	13a_232	13a_396	13a_721
12a_611	12a_1102	12n_632	13a_247	13a_410	13a_730
12a_639	12a_1115	12n_784	13a_253	13a_505	13a_732
12a_698	12a_1208	12n_831	13a_307	13a_518	13a_734
12a_709	12n_32	13a_5	13a_338	13a_530	13a_777
12a_750	12n_70	13a_6	13a_342	13a_538	13a_845

13a_906	13a_2308	13a_3598	13a_4705	13n_1494	13n_3745
13a_916	13a_2323	13a_3609	13a_4744	13n_1582	13n_3752
13a_925	13a_2328	13a_3611	13a_4746	13n_1585	13n_3771
13a_988	13a_2339	13a_3618	13a_4794	13n_1609	13n_3899
13a_995	13a_2389	13a_3653	13a_4848	13n_1628	13n_3902
13a_1027	13a_2411	13a_3657	13n_7	13n_1638	13n_3916
13a_1068	13a_2414	13a_3692	13n_30	13n_1678	13n_3919
13a_1072	13a_2502	13a_3741	13n_50	13n_1688	13n_3921
13a_1077	13a_2505	13a_3758	13n_54	13n_1693	13n_3923
13a_1103	13a_2560	13a_3764	13n_78	13n_1726	13n_3933
13a_1106	13a_2595	13a_3772	13n_149	13n_1789	13n_4023
13a_1122	13a_2645	13a_3776	13n_153	13n_1932	13n_4030
13a_1134	13a_2788	13a_3786	13n_157	13n_1935	13n_4107
13a_1163	13a_2828	13a_3787	13n_231	13n_1992	13n_4115
13a_1169	13a_2870	13a_3805	13n_290	13n_2022	13n_4152
13a_1176	13a_2925	13a_3807	13n_330	13n_2080	13n_4155
13a_1186	13a_3012	13a_3811	13n_341	13n_2193	13n_4162
13a_1193	13a_3028	13a_3905	13n_347	13n_2265	13n_4193
13a_1230	13a_3039	13a_3928	13n_360	13n_2383	13n_4202
13a_1231	13a_3046	13a_3955	13n_370	13n_2404	13n_4247
13a_1237	13a_3057	13a_3993	13n_406	13n_2406	13n_4300
13a_1250	13a_3058	13a_4077	13n_418	13n_2453	13n_4305
13a_1277	13a_3167	13a_4103	13n_428	13n_2503	13n_4313
13a_1316	13a_3181	13a_4152	13n_529	13n_2610	13n_4333
13a_1324	13a_3188	13a_4195	13n_530	13n_2629	13n_4334
13a_1362	13a_3190	13a_4198	13n_561	13n_2694	13n_4351
13a_1380	13a_3242	13a_4211	13n_563	13n_2706	13n_4354
13a_1421	13a_3258	13a_4220	13n_620	13n_2741	13n_4380
13a_1436	13a_3266	13a_4247	13n_671	13n_2765	13n_4457
13a_1438	13a_3274	13a_4268	13n_813	13n_2812	13n_4477
13a_1454	13a_3275	13a_4365	13n_816	13n_2834	13n_4509
13a_1511	13a_3279	13a_4367	13n_839	13n_2837	13n_4515
13a_1514	13a_3302	13a_4398	13n_852	13n_2851	13n_4521
13a_1532	13a_3306	13a_4403	13n_855	13n_2857	13n_4614
13a_1598	13a_3333	13a_4412	13n_892	13n_2926	13n_4632
13a_1608	13a_3343	13a_4419	13n_932	13n_2942	13n_4642
13a_1609	13a_3355	13a_4437	13n_988	13n_2966	13n_4666
13a_1624	13a_3356	13a_4443	13n_994	13n_2977	13n_4709
13a_1648	13a_3403	13a_4475	13n_999	13n_2985	13n_4746
13a_1656	13a_3408	13a_4487	13n_1057	13n_3108	13n_4870
13a_1666	13a_3415	13a_4493	13n_1061	13n_3190	13n_4896
13a_1671	13a_3418	13a_4513	13n_1062	13n_3198	13n_4940
13a_1712	13a_3462	13a_4524	13n_1067	13n_3236	13n_5006
13a_1719	13a_3463	13a_4544	13n_1114	13n_3311	13n_5009
13a_1722	13a_3475	13a_4577	13n_1201	13n_3344	13n_5012
13a_1727	13a_3477	13a_4591	13n_1243	13n_3351	13n_5015
13a_1756	13a_3494	13a_4603	13n_1275	13n_3452	13n_5041
13a_1758	13a_3505	13a_4629	13n_1302	13n_3526	13n_5089
13a_1862	13a_3515	13a_4647	13n_1328	13n_3641	13n_5095
13a_2094	13a_3522	13a_4648	13n_1419	13n_3667	13n_5104
13a_2166	13a_3556	13a_4696	13n_1490	13n_3695	13n_5108
13a_2218	13a_3579	13a_4697	13n_1492	13n_3740	

$\gamma_4 = 2$ **(1565 knots).**

11a_17	12a_186	12a_762	12a_1097	12n_402	13a_89
11a_20	12a_188	12a_763	12a_1099	12n_432	13a_97
11a_49	12a_193	12a_767	12a_1103	12n_472	13a_98
11a_51	12a_224	12a_783	12a_1107	12n_503	13a_103
11a_53	12a_254	12a_787	12a_1112	12n_509	13a_112
11a_64	12a_264	12a_792	12a_1114	12n_517	13a_122
11a_89	12a_276	12a_795	12a_1116	12n_542	13a_134
11a_144	12a_289	12a_796	12a_1120	12n_557	13a_150
11a_161	12a_296	12a_806	12a_1122	12n_585	13a_151
11a_163	12a_297	12a_811	12a_1137	12n_603	13a_154
11a_178	12a_338	12a_812	12a_1143	12n_627	13a_157
11a_192	12a_347	12a_813	12a_1152	12n_638	13a_160
11a_214	12a_370	12a_815	12a_1164	12n_639	13a_162
11a_240	12a_371	12a_816	12a_1172	12n_671	13a_170
11a_241	12a_392	12a_817	12a_1177	12n_672	13a_171
11a_248	12a_428	12a_822	12a_1184	12n_675	13a_185
11a_274	12a_437	12a_828	12a_1193	12n_682	13a_201
11a_278	12a_449	12a_841	12a_1197	12n_688	13a_203
11a_292	12a_460	12a_844	12a_1200	12n_691	13a_209
11a_293	12a_461	12a_846	12a_1222	12n_692	13a_214
11a_299	12a_480	12a_869	12a_1223	12n_694	13a_217
11a_319	12a_507	12a_873	12a_1229	12n_707	13a_225
11a_329	12a_508	12a_876	12a_1232	12n_712	13a_231
11a_338	12a_514	12a_880	12a_1240	12n_717	13a_236
11a_339	12a_523	12a_881	12a_1244	12n_722	13a_251
11a_354	12a_533	12a_885	12a_1248	12n_734	13a_257
11a_355	12a_542	12a_890	12a_1252	12n_743	13a_273
11a_360	12a_546	12a_899	12a_1255	12n_746	13a_275
11a_362	12a_553	12a_905	12a_1265	12n_755	13a_284
11a_366	12a_556	12a_912	12a_1270	12n_788	13a_295
12a_9	12a_570	12a_950	12a_1274	12n_796	13a_304
12a_29	12a_575	12a_974	12n_27	12n_803	13a_309
12a_35	12a_592	12a_986	12n_33	12n_805	13a_314
12a_39	12a_597	12a_994	12n_52	12n_818	13a_333
12a_43	12a_610	12a_996	12n_67	12n_820	13a_334
12a_47	12a_616	12a_1001	12n_104	12n_827	13a_345
12a_49	12a_628	12a_1002	12n_107	12n_830	13a_355
12a_52	12a_633	12a_1006	12n_148	12n_863	13a_362
12a_55	12a_636	12a_1009	12n_153	12n_866	13a_366
12a_64	12a_641	12a_1013	12n_164	12n_874	13a_370
12a_66	12a_650	12a_1021	12n_169	12n_880	13a_374
12a_67	12a_651	12a_1025	12n_173	13a_4	13a_393
12a_76	12a_653	12a_1030	12n_174	13a_8	13a_402
12a_85	12a_655	12a_1031	12n_185	13a_16	13a_403
12a_88	12a_657	12a_1035	12n_186	13a_25	13a_406
12a_96	12a_661	12a_1036	12n_190	13a_29	13a_408
12a_97	12a_665	12a_1037	12n_211	13a_35	13a_423
12a_103	12a_674	12a_1044	12n_229	13a_51	13a_426
12a_113	12a_686	12a_1049	12n_243	13a_52	13a_429
12a_114	12a_689	12a_1052	12n_247	13a_56	13a_433
12a_117	12a_692	12a_1058	12n_260	13a_59	13a_437
12a_136	12a_702	12a_1059	12n_269	13a_65	13a_438
12a_137	12a_703	12a_1079	12n_289	13a_71	13a_443
12a_143	12a_706	12a_1080	12n_305	13a_82	13a_451
12a_146	12a_730	12a_1084	12n_317	13a_84	13a_468
12a_152	12a_738	12a_1088	12n_330	13a_85	13a_473
12a_167	12a_753	12a_1091	12n_361	13a_88	13a_474

13a_480	13a_826	13a_1377	13a_1820	13a_2245	13a_2593
13a_481	13a_866	13a_1378	13a_1825	13a_2249	13a_2598
13a_496	13a_867	13a_1381	13a_1827	13a_2255	13a_2600
13a_497	13a_895	13a_1396	13a_1854	13a_2263	13a_2604
13a_508	13a_900	13a_1400	13a_1859	13a_2265	13a_2606
13a_513	13a_910	13a_1404	13a_1864	13a_2268	13a_2607
13a_525	13a_918	13a_1405	13a_1865	13a_2269	13a_2610
13a_535	13a_924	13a_1406	13a_1867	13a_2270	13a_2611
13a_536	13a_931	13a_1414	13a_1895	13a_2284	13a_2616
13a_539	13a_933	13a_1419	13a_1901	13a_2286	13a_2628
13a_540	13a_955	13a_1440	13a_1922	13a_2287	13a_2636
13a_542	13a_964	13a_1442	13a_1927	13a_2290	13a_2641
13a_544	13a_966	13a_1452	13a_1928	13a_2310	13a_2644
13a_545	13a_976	13a_1461	13a_1943	13a_2333	13a_2655
13a_546	13a_980	13a_1464	13a_1946	13a_2335	13a_2656
13a_551	13a_991	13a_1465	13a_1952	13a_2359	13a_2679
13a_564	13a_1002	13a_1466	13a_1955	13a_2360	13a_2683
13a_569	13a_1020	13a_1472	13a_1957	13a_2370	13a_2697
13a_571	13a_1041	13a_1475	13a_1960	13a_2372	13a_2706
13a_573	13a_1045	13a_1495	13a_1961	13a_2375	13a_2709
13a_574	13a_1050	13a_1503	13a_1972	13a_2376	13a_2711
13a_584	13a_1051	13a_1524	13a_1982	13a_2386	13a_2724
13a_589	13a_1063	13a_1541	13a_1984	13a_2387	13a_2732
13a_593	13a_1066	13a_1542	13a_1985	13a_2390	13a_2735
13a_594	13a_1069	13a_1543	13a_1987	13a_2395	13a_2751
13a_598	13a_1070	13a_1544	13a_1989	13a_2399	13a_2757
13a_599	13a_1099	13a_1546	13a_1997	13a_2400	13a_2761
13a_603	13a_1102	13a_1547	13a_1999	13a_2405	13a_2763
13a_606	13a_1120	13a_1550	13a_2018	13a_2407	13a_2767
13a_618	13a_1129	13a_1553	13a_2025	13a_2408	13a_2769
13a_620	13a_1137	13a_1566	13a_2030	13a_2409	13a_2772
13a_621	13a_1142	13a_1567	13a_2032	13a_2419	13a_2782
13a_622	13a_1147	13a_1580	13a_2037	13a_2421	13a_2784
13a_623	13a_1149	13a_1583	13a_2042	13a_2428	13a_2786
13a_625	13a_1158	13a_1590	13a_2052	13a_2434	13a_2790
13a_629	13a_1182	13a_1603	13a_2054	13a_2438	13a_2813
13a_631	13a_1183	13a_1614	13a_2055	13a_2443	13a_2814
13a_636	13a_1188	13a_1618	13a_2064	13a_2450	13a_2816
13a_643	13a_1192	13a_1621	13a_2076	13a_2452	13a_2821
13a_644	13a_1203	13a_1629	13a_2077	13a_2453	13a_2823
13a_656	13a_1233	13a_1630	13a_2078	13a_2462	13a_2829
13a_658	13a_1235	13a_1634	13a_2081	13a_2474	13a_2840
13a_661	13a_1245	13a_1641	13a_2088	13a_2476	13a_2843
13a_669	13a_1246	13a_1643	13a_2092	13a_2478	13a_2853
13a_679	13a_1255	13a_1652	13a_2093	13a_2485	13a_2856
13a_682	13a_1258	13a_1660	13a_2098	13a_2520	13a_2863
13a_686	13a_1259	13a_1683	13a_2107	13a_2526	13a_2865
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